

```

R1(config)#int e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no
R1(config-if)#no sh
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no shut
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#w
R1(config)#wri
R1(config)#exit
R1#
*Aug 26 09:54:58.293: %SYS-5-CONFIG_I: Configured from console by console
R1#write
Building configuration...
[OK]
R1#con
R1#conf
R1#configure t
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#write
^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#wri
*Aug 26 09:55:34.983: %SYS-5-CONFIG_I: Configured from console by console
R1#write
Building configuration...
[OK]
R1#
*Aug 26 09:56:43.356: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	20.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R1#

```

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2

```

--More--

```

R2#
R2#conf
R2#configure ter
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int 0/0
      ^
% Invalid input detected at '^' marker.

R2(config)#int e0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no sh
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int e0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no
R2(config-if)#no sh
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#write
      ^
% Invalid input detected at '^' marker.

R2(config)#exit
R2#wr
*Aug 26 09:58:35.898: %SYS-5-CONFIG_I: Configured from console by console
R2#write
Building configuration...
[OK]
R2#
*Aug 26 10:01:44.188: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2#
R2#
R2#
R2#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	20.0.0.2	YES	manual	up	up
Ethernet0/1	30.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R2#

```

```

R2#write
Building configuration...
[OK]
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     20.0.0.0/8 is directly connected, Ethernet0/0
L     20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/1
L     30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2

```

```

R3#
R3#conf t
R3#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int e0/0
R3(config-if)#ip add 30.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#int e0/1
R3(config-if)#ip add 172.19.0.1 255.255.0.0
R3(config-if)#no sh
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#int e0/2
R3(config-if)#ip add 21.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Aug 26 10:04:45.668: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3(config)#write
^
% Invalid input detected at '^' marker.

R3(config)#exit
R3#writ
*Aug 26 10:05:11.158: %SYS-5-CONFIG_I: Configured from console by console
R3#write
Building configuration...
[OK]
R3#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	30.0.0.2	YES	manual	up	up
Ethernet0/1	172.19.0.1	YES	manual	up	up
Ethernet0/2	21.0.0.1	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R3#

```

```

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/2
L     21.0.0.1/32 is directly connected, Ethernet0/2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/0
L     30.0.0.2/32 is directly connected, Ethernet0/0
S    168.14.0.0/16 [1/0] via 21.0.0.2
     172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.19.0.0/16 is directly connected, Ethernet0/1
L     172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
R3#

```

```

R4#
R4#conf t
R4#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int e0/1
R4(config-if)#ip add 172.19.0.2 255.255.0.0
R4(config-if)#no sh
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#exit
R4#
*Aug 26 10:07:08.735: %SYS-5-CONFIG_I: Configured from console by console
R4#write
Building configuration...
[OK]
R4#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	unassigned	YES	TFTP	up	up
Ethernet0/1	172.19.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R4#conf
R4#configure t
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int e0/0
R4(config-if)#ip add 15.0.0.1 255.0.0.0
R4(config-if)#no sh
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#exit
R4#wri
*Aug 26 10:08:54.381: %SYS-5-CONFIG_I: Configured from console by console
R4#write
Building configuration...
[OK]
R4#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	15.0.0.1	YES	manual	up	up
Ethernet0/1	172.19.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R4#

```

```

R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
R4#

```

```

R5#
R5#
R5#conf
R5#configure ter
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#
R5(config)#int e0/0
R5(config-if)#ip add 21.0.0.2 255.0.0.0
R5(config-if)#no ssh
R5(config-if)#no sh
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#int e0/1
R5(config-if)#ip add 192.168.168.1 255.255.255.0
R5(config-if)#no sh
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#int e0/1
R5(config-if)#
R5(config-if)#
R5(config-if)#ip add 192.168.168.1 255.255.255.0
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#exit
R5#wri
*Aug 26 10:16:32.082: %SYS-5-CONFIG_I: Configured from console by console
R5#write
Building configuration...
[OK]
R5#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	21.0.0.2	YES	manual	up	up
Ethernet0/1	192.168.168.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R5#

```

```

R5#show ip rout
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/0
L     21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
     192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
L     192.168.168.1/32 is directly connected, Ethernet0/1
R5#

```

```

R6#
R6#conf
R6#configure ter
R6#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#int e0/0
R6(config-if)#ip add 162.14.0.1 255.255.0.0
R6(config-if)#no shou
R6(config-if)#no shou
R6(config-if)#no shout
R6(config-if)#no shu
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#int e0/1
R6(config-if)#ip add 192.168.168.1 255.255.255.0
R6(config-if)#
*Aug 26 10:20:43.973: %IP-4-DUPADDR: Duplicate address 192.168.168.1 on Ethernet0/1, sourced by aabb.cc00.0510
R6(config-if)#no sh
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#write
R6(config)#write
R6(config)#exit
R6#wri
*Aug 26 10:21:09.022: %SYS-5-CONFIG_I: Configured from console by console
R6#write
Building configuration...
[OK]
R6#sho
*Aug 26 10:21:14.194: %IP-4-DUPADDR: Duplicate address 192.168.168.1 on Ethernet0/1, sourced by aabb.cc00.0510
R6#show ip int ter
R6#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	162.14.0.1	YES	manual	up	up
Ethernet0/1	192.168.168.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R6#

```

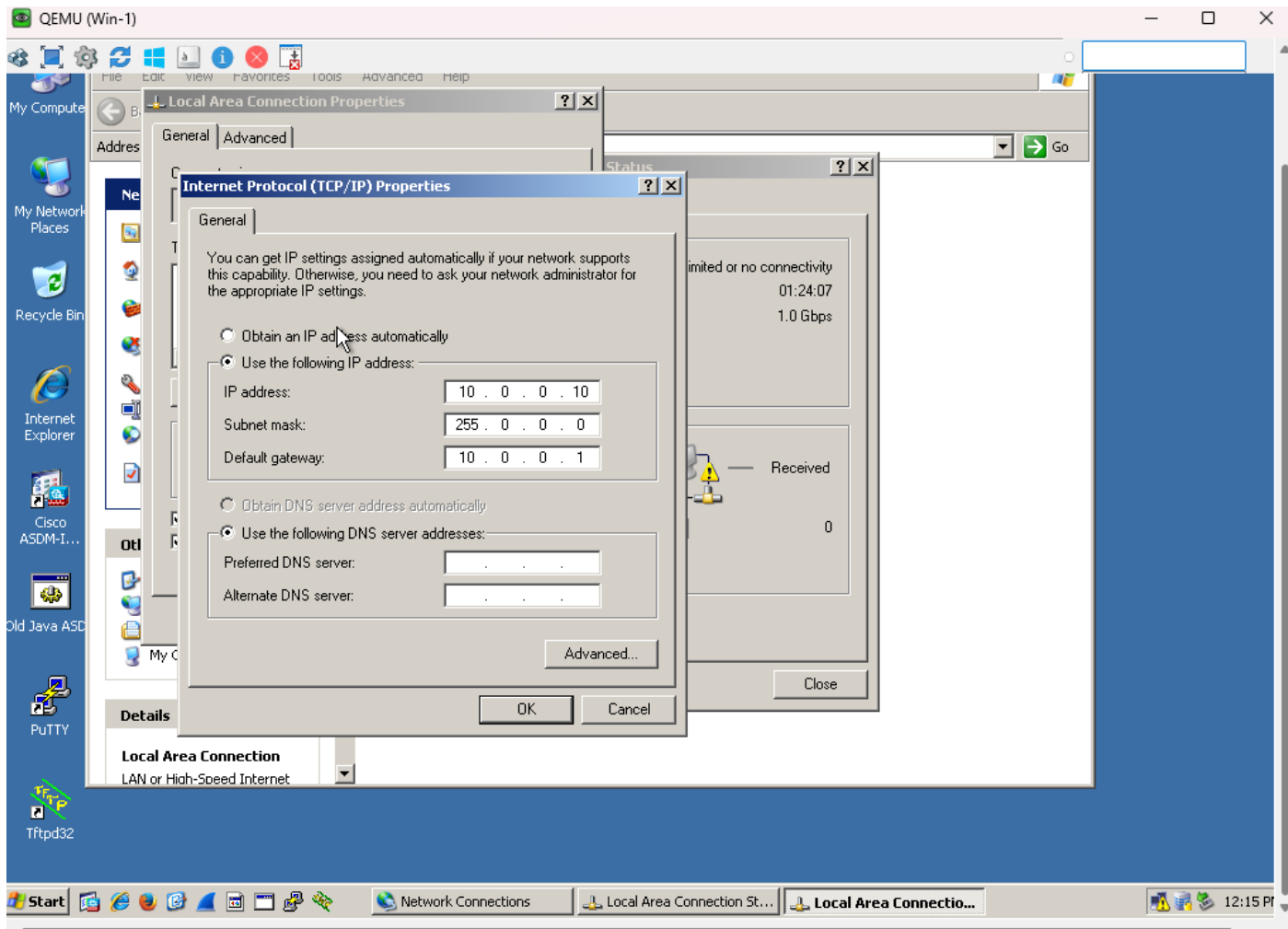
```

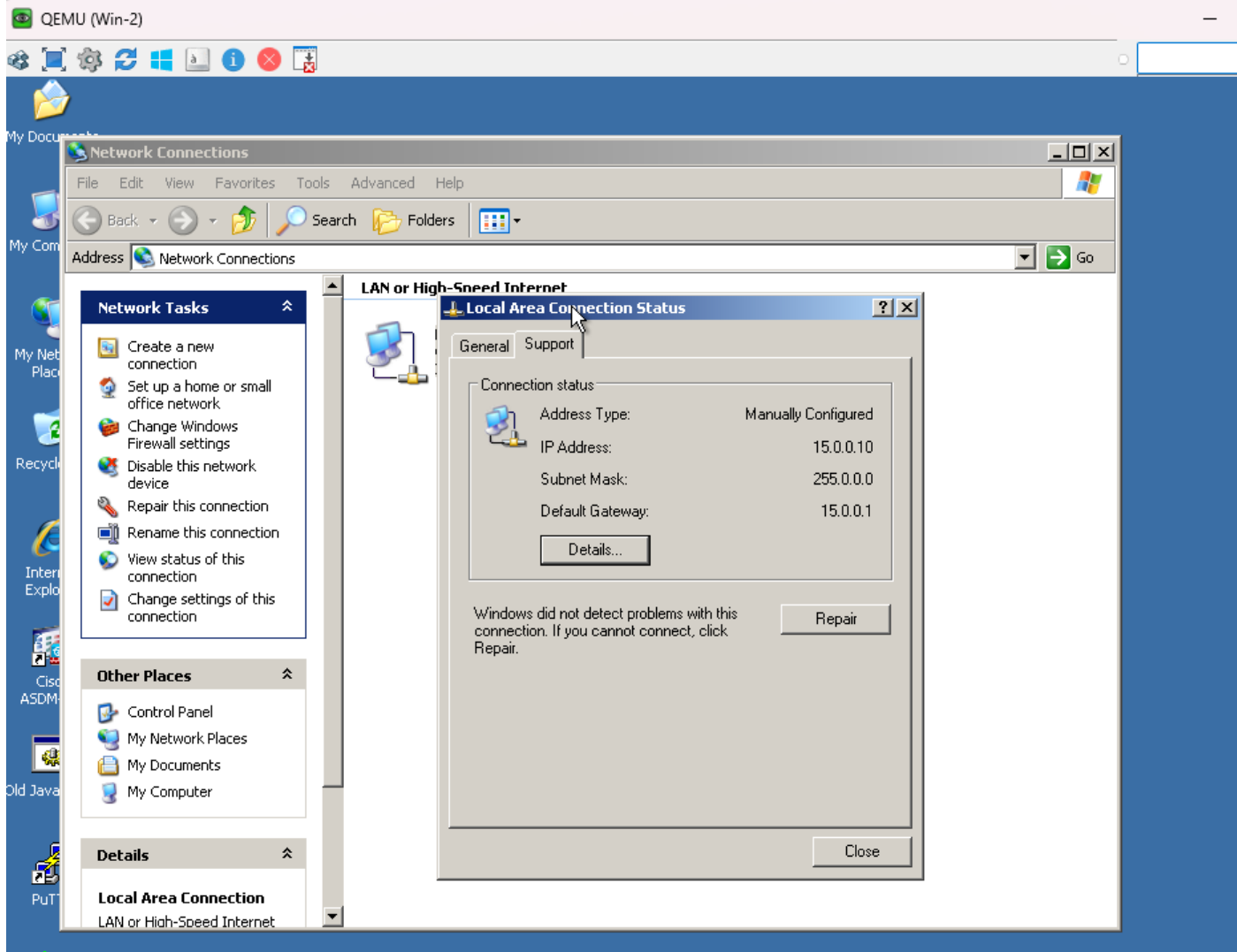
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

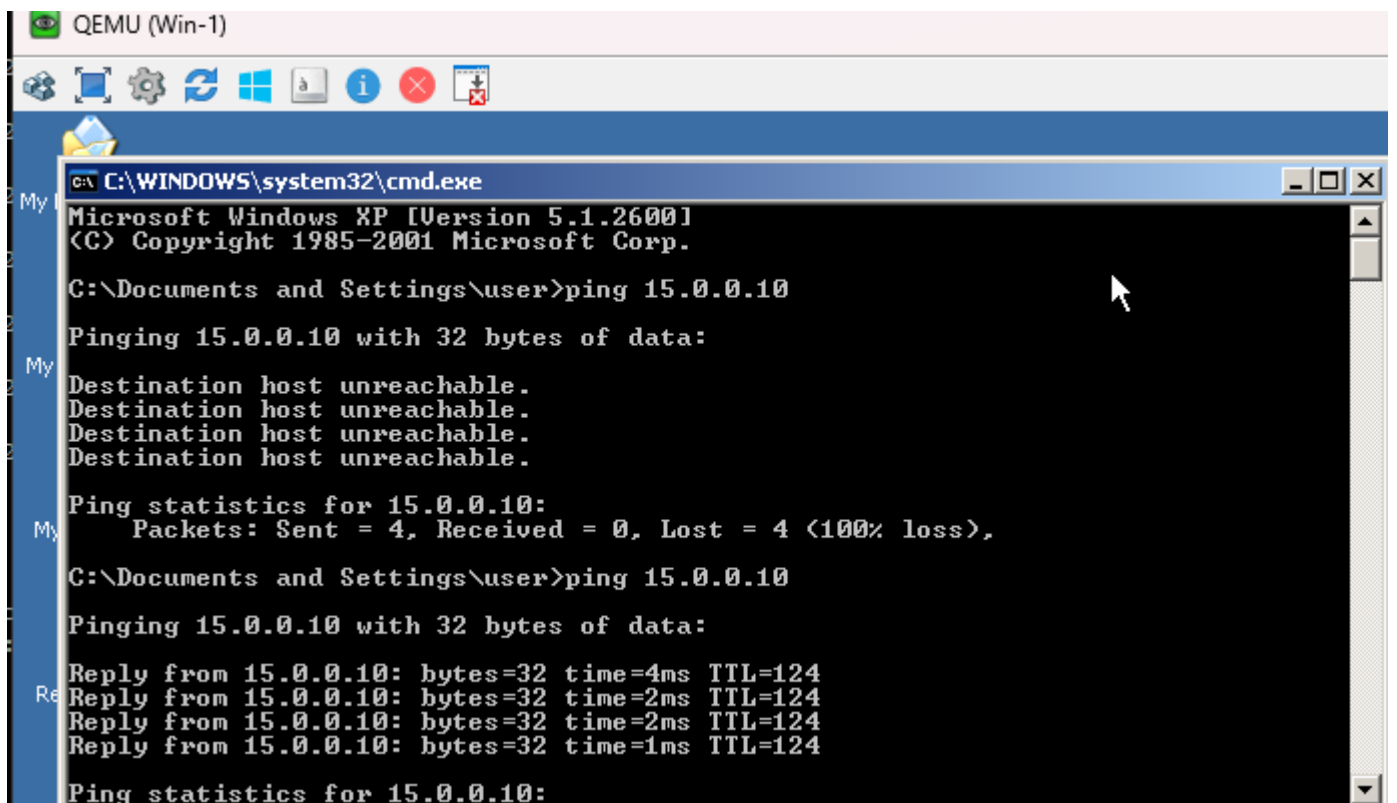
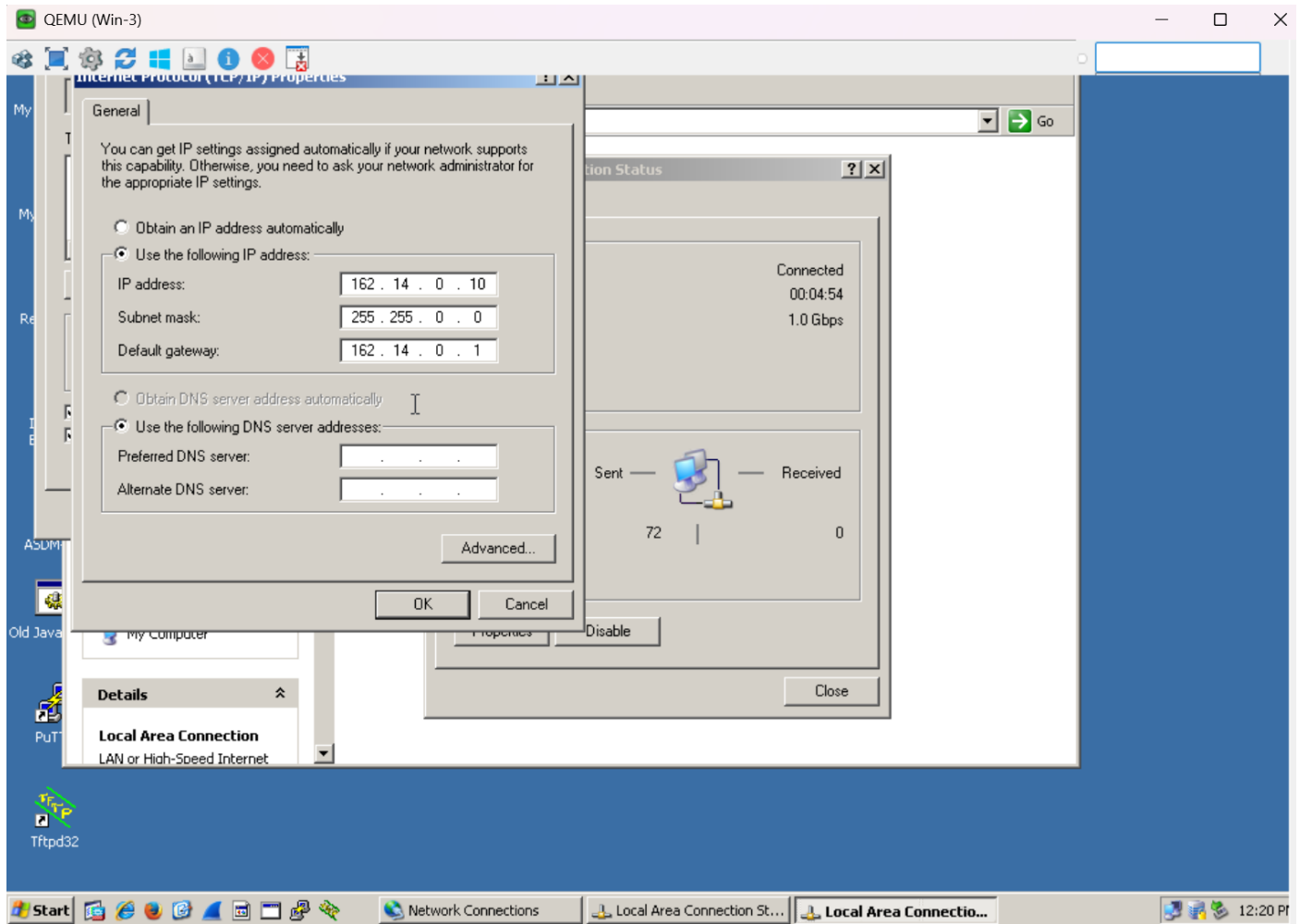
Gateway of last resort is not set

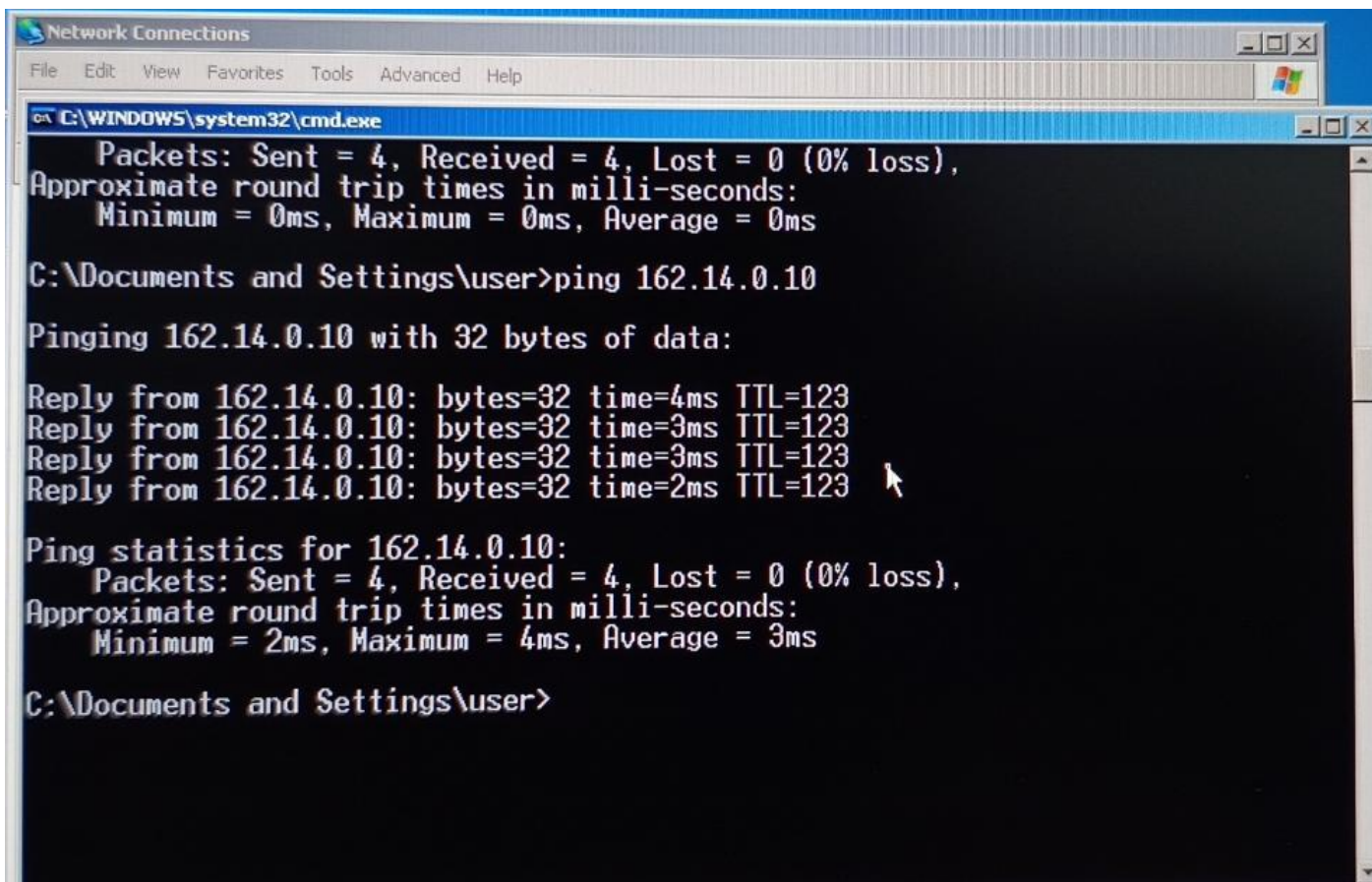
S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
      C    162.14.0.0/16 is directly connected, Ethernet0/0
      L    162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
      C    192.168.168.0/24 is directly connected, Ethernet0/1
      L    192.168.168.2/32 is directly connected, Ethernet0/1
R6#

```









The image shows a Windows 'Network Connections' window. Inside, a command prompt (cmd.exe) is open at the path 'C:\WINDOWS\system32\cmd.exe'. The prompt shows the output of a 'ping' command to the IP address 162.14.0.10. The output indicates that 4 packets were sent and received with 0% loss. The round trip times are listed as Minimum = 0ms, Maximum = 0ms, and Average = 0ms. Below this, the command 'C:\Documents and Settings\user>ping 162.14.0.10' is entered. The response shows 'Pinging 162.14.0.10 with 32 bytes of data:' followed by four replies from 162.14.0.10, each with 32 bytes, times of 4ms, 3ms, 3ms, and 2ms, and a TTL of 123. Finally, 'Ping statistics for 162.14.0.10:' are shown, with 4 packets sent and received, 0% loss, and round trip times of Minimum = 2ms, Maximum = 4ms, and Average = 3ms. The prompt ends with 'C:\Documents and Settings\user>'.

```
Network Connections
File Edit View Favorites Tools Advanced Help
C:\WINDOWS\system32\cmd.exe
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=4ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>
```

```
QEMU (Win-2)
C:\Documents and Settings\user>ping 162.14.0.10
Pinging 162.14.0.10 with 32 bytes of data:
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
C:\Documents and Settings\user>
```

```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>
```